

SECOND EDITION

Biogeochemistry of Wetlands

Science and Applications

**K. Ramesh Reddy
Ronald D. DeLaune
Patrick W. Inglett**



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Biogeochemistry of Wetlands

The globally important nature of wetland ecosystems has led to their increased protection and restoration as well as their use in engineered systems. Underpinning the beneficial functions of wetlands are a unique suite of physical, chemical, and biological processes that regulate elemental cycling in soils and the water column. This book provides an in-depth coverage of these wetland biogeochemical processes related to the cycling of macroelements, including carbon, nitrogen, phosphorus, and sulfur; secondary and trace elements; and toxic organic compounds.

In this synthesis, the authors combine more than 100 years of experience studying wetlands and biogeochemistry to look inside the black box of elemental transformations in wetland ecosystems. This new edition is updated throughout to include more topics and provide an integrated view of the coupled nature of biogeochemical cycles in wetland systems. The influence of the elemental cycles is discussed at a range of scales in the context of environmental change including climate, sea level rise, and water quality. Frequent examples of key methods and major case studies are also included to help the reader extend the basic theories for application in their own system. Some of the major topics discussed are:

- Flooded soil and sediment characteristics
- Aerobic-anaerobic interfaces
- Redox chemistry in flooded soil and sediment systems
- Anaerobic microbial metabolism
- Plant adaptations to reducing conditions
- Regulators of organic matter decomposition and accretion
- Major nutrient sources and sinks
- Greenhouse gas production and emission
- Elemental flux processes
- Remediation of contaminated soils and sediments
- Coupled C-N-P-S processes
- Consequences of environmental change in wetlands

The book provides the foundation for a basic understanding of key biogeochemical processes and its applications to solve real-world problems. It is detailed, but also assists the reader with box inserts, artfully designed diagrams, and summary tables all supported by numerous current references. This book is an excellent resource for senior undergraduates and graduate students studying ecosystem biogeochemistry with a focus in wetlands and aquatic systems.



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Preface

Wetland science has now emerged as a discipline where hydrologists, biogeochemists, pedologists, ecologists, microbiologists, and scientists from various other disciplines are working individually and together to improve our understanding of the functions and ecosystem services of wetlands. The idea for this book came during the early 1980s when the first author began teaching a new course entitled “Biogeochemistry of Wetlands” at the University of Florida. A similar course was also taught at Louisiana State University during the early 1970s under the title of “Chemistry of Flooded Soils,” and later in the 1980s the course title was changed to “Biogeochemistry of Wetland Soils and Sediments.” Requests for a book were constant by the 1000+ students who took this course before the first edition of the book was published in 2008. A similar interest to have a wetland biogeochemistry reference book was also expressed by colleagues working with other universities, governmental agencies, and other organizations.

The 2008 edition of the book primarily focused on the role of biogeochemistry as the key “operating system” that regulates the physical, chemical, and biological processes of elemental cycles within a wetland, thereby affecting large-scale ecosystem services. We focused on “organic matter” as a hub of biogeochemistry and oxidation–reduction reactions as primary drivers of biogeochemical processes. Wetlands are unique in that a range of soil–sediment conditions, from strongly reducing (anaerobic) to oxidizing (aerobic), can be found at a range of spatial and temporal scales. These environments include a wide range of systems such as tidal freshwater and salt marshes, inland freshwater marshes and northern peat lands, swamp and bottomland forests, and riparian wetlands and estuaries. In certain topical areas, we relied heavily on the biogeochemistry of aquatic systems because a limited amount of information was available on wetlands.

The first edition was written as both a reference and a text for a graduate-level course. Individuals with an interest in environmental science, biology, chemistry, ecology, and environmental engineering would also find this book useful. The book is highly cited by researchers from various disciplines. The impact of soil redox processes on elemental cycling, biotransformation, and heavy metal chemistry was emphasized. The book includes chapters dealing with terminology and electrochemical properties describing basic biogeochemical processes that drive transformation processes in wetlands. Detailed chapters included carbon, oxygen, nitrogen, phosphorus, sulfur, iron and manganese, metals and metalloids, and toxic organics. In other chapters, we discussed plant adaptation to wetland conditions, climate change, greenhouse gas emissions, and case studies where extensive advancements have been made.

We wrote the second edition because the revisions to the first edition were long overdue. Our revision was based on the feedback from colleagues and from the many University of Florida students who used the first edition book in the “Biogeochemistry of Wetlands” course. As a result of this feedback, many corrections and changes were made in the second edition. First and foremost, we added a third author (Patrick Inglett) to the book to have an impact and thinking of the next generation of wetland science experts. In the second edition, we added some new concepts and helped streamline some of the topic areas for better application to emerging areas of research. During revisions, we removed the last five chapters from the first edition and reincorporated some of those key messages into other chapters. The second edition is also significantly improved with the addition of many new references, special topic boxes to improve the teaching quality, and all images are now in color.

The focus of the second edition was to provide an integrative approach to wetland biogeochemical cycles and the coupling and decoupling of macrolelements (carbon, nitrogen, phosphorus, and sulfur) as related to climate change, sea level rise, and water quality. We added a new synthesis chapter that presents an integrative approach to coupled biogeochemical cycles and its environmental and ecological significance. We sincerely hope the second edition provides state-of-the-art

scientific information involving wetland and aquatic systems biogeochemistry. As always, when we wrote this book, we learned even more and made our best attempt to convey a contemporary message on all aspects of wetland biogeochemistry. For those who read this book as a student, or as an instructor, or others using it as reference material, we appreciate your comments and feedback to help guide the next edition (planned in the next three to four years). We hope this book builds on the first edition and continues to unite scientists of all disciplines in the study of wetlands and serves as a foundation for building and inspiring the next generation of wetland professionals.

Acknowledgments

We extend our sincere thanks to many of our colleagues who reviewed the content of the first edition, provided information to support the chapters, and helped to improve the quality of the book. These include M. Clark, R. Corstanje, E. D'Angelo, W. F. DeBusk, M. Fisher, W. Hurt (*deceased*), R. Gambrell, K. S. Inglett, J. Jawitz, T. Osborne, J. White, and many graduate students, postdoctoral fellows, and visiting scientists of the Wetland Biogeochemistry Laboratory at the University of Florida. Many illustrations presented in this book were originally developed as part of a graduate course taught at the University of Florida, but the first two authors appreciate Patrick W. Inglett for his efforts to create, update, and redraw the figures in both the first and second editions. All three of us sincerely appreciate the contributions of those who came before us (on whose shoulders we stand), as well as the creativity and hard work of the many students and post-docs we have advised. Lastly, we gratefully acknowledge the support and love of our spouses (Sulochana Reddy, Carole DeLaune [*deceased*], and Kanika Inglett) and families, without which this type of effort simply would not be possible, or nearly as rewarding.

K. Ramesh Reddy

University of Florida

Institute of Food and Agricultural Sciences

School of Natural Resources and Environment &

Wetland Biogeochemistry Laboratory

Soil, Water and Ecosystem Sciences Department

Gainesville, Florida

Ronald D. DeLaune

Louisiana State University

Wetland Biogeochemistry Institute

Department of Oceanography and Coastal Sciences

School of the Coast and Environment

Baton Rouge, Louisiana

Patrick W. Inglett

University of Florida

Institute of Food and Agricultural Sciences

Wetland Biogeochemistry Laboratory

Soil, Water and Ecosystem Sciences Department

Gainesville, Florida



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Authors

K. Ramesh Reddy is a graduate research professor and director (2019 to present) of the School of Natural Resources and Environment at the University of Florida (UF). Dr. Reddy served as chair (2000–2018) of the UF Soil and Water Sciences Department. Dr. Reddy carried out research for five decades on biogeochemical cycling of nutrients in natural and managed ecosystems as related to water quality, carbon and nutrient sequestration, and greenhouse gas emission. Dr. Reddy taught the “Biogeochemistry of Wetlands” course at UF for 40+ years. Dr. Reddy published 420+ refereed journal articles and book chapters and edited five books. During his tenure at UF, Dr. Reddy served as major advisor for 34 PhD and 20 MS students and served on 120+ graduate student committees. Dr. Reddy has served on numerous advisory committees at state, national, and international levels. Dr. Reddy’s select awards and honors include 1993: appointed as UF Graduate Research Professor (distinguished professorship); UF Doctoral Dissertation Advisory/Mentoring Award; Environmental Quality Research Award, American Society of Agronomy; Soil Science Applied Research Award, Soil Science Society of America; Fellow, American Association for the Advancement of Science; Fellow–Soil Science Society of America; Fellow–American Society of Agronomy; 2012 Lifetime Achievement Award–INTECOL–Wetlands; 2016 National Wetlands Award–Research–Environmental Law Institute, Washington DC; and 2016 Lifetime Achievement Award–Society of Wetland Scientists. Dr. Reddy obtained a BS and MS from A.P. Agricultural University, India, and a PhD (1976) from Louisiana State University.

Ronald D. DeLaune is an emeritus research professor in the Wetland Biogeochemistry Group, Department of Oceanography and Coastal Sciences, School of the Coast and Environment, Louisiana State University, Baton Rouge, Louisiana. His research interests include nutrient cycling in agriculture and wetland ecosystems, role of wetlands in global biogeochemical cycles, wetland soil–plant interactions, coastal restoration and processes governing marsh stability, redox chemistry of flooded soil and sediment, water quality of aquatic environments, greenhouse gas emissions from wetlands, and the fate of organic pollutants and toxic heavy metals entering wetlands. Dr. DeLaune has served as associate editor or on the editorial board of numerous scientific journals. He has authored more than 350 scientific papers in peer-reviewed journals. Dr. DeLaune is also a fellow in the Soil Science Society of America. Dr. DeLaune received his BS and MS from Louisiana State University, and PhD from University of Wageningen, The Netherlands.

Patrick W. Inglett first became involved in wetlands when he worked in the Wetlands Protection Section at the U.S. Environmental Protection Agency in Atlanta, Georgia. His interest was the science behind the conservation and regulatory process for these unique ecosystems. He received his BS in applied biology with a minor in earth and atmospheric sciences from the Georgia Institute of Technology, and later, his MS and PhD from the University of Florida. Dr. Inglett is currently a professor of biogeochemistry in the Soil and Water Sciences Department at the University of Florida, where he also serves as the director of the Wetland Biogeochemistry Laboratory core analytical and research facility. His research interests include nutrient cycling processes related to ecosystem function and restoration, particularly aquatic systems. He specializes in nitrogen and phosphorus limitation, enzymes, nitrogen fixation, temperature sensitivity of microbial processes, greenhouse gas production, and the use of stable isotopes. For the last 15 years, Dr. Inglett has taught both undergraduate and graduate courses in Environmental Biogeochemistry, Advanced Biogeochemistry, and on occasion, the Wetlands Biogeochemistry course. He has advised more than 50 graduate and undergraduate students, post-docs, and visiting scholars, and he is the author of more than 60 journal articles and book chapters.



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1 Introduction

Wetlands are unique ecosystems located in areas with comparatively low elevation and a high water table. Wetlands can include marshes, swamps, bogs, and similar areas that are poorly drained and retain water during rainy periods. Globally, wetlands can be found in all climates, from tropical to tundra, with the exception of Antarctica. Approximately 6% of Earth's land surface, which equals approximately 1,200 million ha., is covered by wetlands (Ramsar Convention on Wetlands, 2018; Davidson and Finlayson, 2018). The United States alone accounts for approximately 10% of the global wetlands, or 115 million ha. of wetlands, with Alaska representing approximately 62% of the total wetland area. The conterminous United States accounts for approximately 44 million ha., which includes 95% freshwater wetlands and 5% intertidal coastal wetlands (Dahl, 2011). The Convention on Wetlands, signed in Ramsar, Iran, in 1971, is an intergovernmental treaty that provides the framework for national action and international cooperation for the conservation and wise use of wetlands and their resources. There are presently 158 contracting parties to the convention, with 1,723 wetland sites, totaling 160 million ha., designated for inclusion in the Ramsar List of Wetlands of International Importance (<http://www.ramsar.org/>).

The value and function of wetlands are well recognized, as evidenced by national and international policies to preserve wetland ecosystems. Why are wetlands so worthy of protection? Wetlands are complex ecosystems with functions driven by interactions of many physical, chemical, and biological processes. Wetlands are some of the most biologically productive ecosystems on Earth; their productivity can exceed that of terrestrial and aquatic systems. Wetlands not only serve to promote and sustain biota in many forms but also serve as living filters that process pollutants from terrestrial runoff and atmospheric deposition. Biodegradation of organic compounds, elemental cycling, atmospheric exchange, hydrologic processing and capacity, and plant response are controlled by the unique conditions found in the wetland environment (Figure 1.1).

In the past, scientists often described wetland ecosystems on the basis of their disciplinary bias, with an emphasis on one of the specialties such as hydrology, chemistry, wildlife, microbiology, or vegetation; however, no one discipline or specialization can describe these complex processes. This was like the fabled group of blind men trying to describe an elephant by each touching a different part, such as the side or the trunk or the tail. When the blind men compared their notes, they were in complete disagreement in describing an elephant. The one who had touched the elephant's side claimed an elephant was like a wall; the man who had felt the tail said an elephant was like a rope. It took the whole group of blind men to accurately describe the elephant. Similarly, it has become very clear that no single discipline can adequately describe a complex ecosystem such as wetlands. Instead, describing a wetland ecosystem requires an interdisciplinary approach linking various specializations—biology, biogeochemistry, ecology, environmental science, hydrology, and so on. Understanding must also draw on disciplines outside these fields. For example, much of the chemical and microbiological processes measured in wetland soils is based on studies of saturated soils, which began around the turn of the 20th century with research into nutrient behavior in paddy soils and processes measured in lake and marine sediments.

In landscapes, wetlands typically occur between upland and aquatic ecosystems. Because uplands are often the source of water to wetlands, components within runoff are also supplied from uplands. In the absence of wetlands, contaminants added to or generated within upland areas are directly transported into receiving aquatic ecosystems. Several physical, chemical, and biological processes functioning in the soil of uplands and wetlands are involved in regulating the

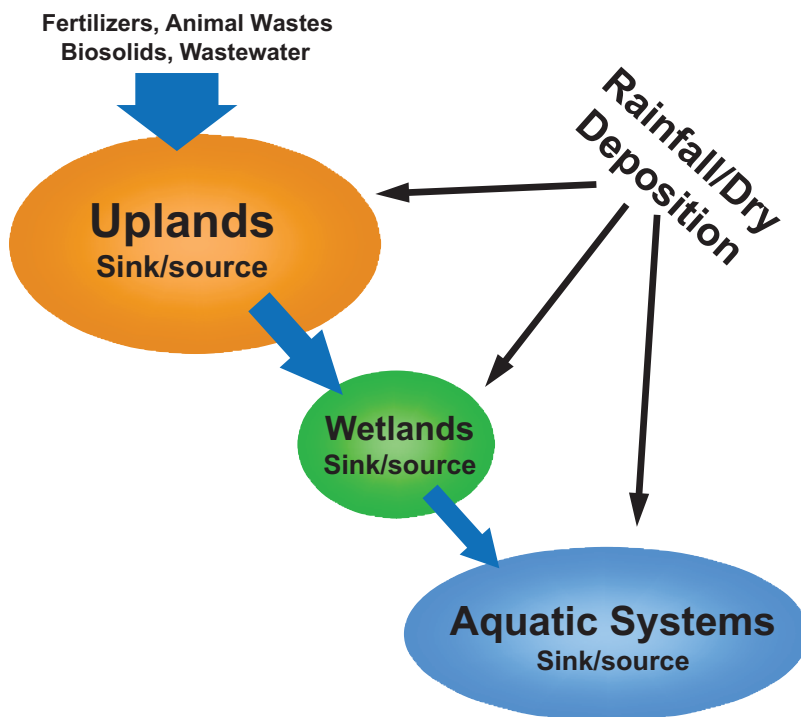


FIGURE 1.1 Linkages between uplands, wetlands, and aquatic systems.

fate (availability) of contaminants. For example, uplands and wetlands can serve as both “sinks,” “sources,” and “transformers” for contaminants:

- *Sink*: Contaminants are transformed into biologically unavailable forms within the system. For example, wetlands can convert nitrate to N_2 gas through a biological reaction called denitrification (this process is discussed in detail in Chapter 8).
- *Source*: Contaminants are transported from one ecosystem to another. For example, uplands can serve as a “source” for suspended solids, nutrients, and other contaminants to wetlands. Similarly, eutrophic wetlands can be a “source” of contaminants or nutrients to adjacent aquatic systems such as streams, rivers, lakes, and estuaries.
- *Transformer*: Contaminants added to a wetland can also be transformed and released as different or complexed compounds, or as new compounds to the aquatic ecosystem downstream. Because wetlands receive runoff from upland ecosystems, the changes in wetlands can be used as an indicator of an upland ecosystem’s “health.”

Because wetland soils can serve as sinks, sources, and transformers of nutrients and other chemical contaminants, they have a significant impact on water quality and ecosystem productivity.

The functions of wetlands in regulating nutrient and elemental storages and transformations are captured in the study of *biogeochemistry*. Biogeochemistry is defined as the study of the exchange or flux of materials between living and nonliving components of the biosphere [Box 1.1]. Like the study of wetlands, biogeochemistry is also an interdisciplinary science, involving the interaction of complex physical, chemical, and biological processes in various components of the ecosystem, including the exchange of materials between biotic and abiotic. As defined, biogeochemistry encompasses interactions from the smallest scale to the global scale encompassing the biosphere. Wetland biogeochemistry, as the focus of this book, principally relates to small-scale exchanges from the particle and microbial scale to the field scale. However, the cumulative impact of these

small-scale processes on a landscape and on global reservoirs can be significant and will also be addressed.

BOX 1.1 A BRIEF HISTORY OF BIOGEOCHEMISTRY

Contrary to most opinions, the idea of biogeochemistry or elemental cycling in earth systems is not a very recent concept. You may be surprised to find that while the term biogeochemistry was first coined only in the early 20th century, the concept and foundation for this discipline are found centuries earlier.

The roots of biogeochemistry began with the strong need for scientific achievement in food production and agriculture. Many of the early scientists were also pioneers of science itself, and nature, soils, and plants were often the subject matter for their basic physics and chemical theories. As early as 1720, the derivation of materials for living biota were known to come from gaseous and soil components, and the pioneer of chemistry, **Lavoisier**, correctly found that according to conservation of mass, the gaseous material required for living plants (carbon dioxide) was derived from the decomposition of plant biomass. In essence, this is the first documented biogeochemical cycle. Later work by **von Liebig** and others properly related the role of soil solution in supplying other water and particular or limiting nutrients to plant growth. Liebig's work was so important that his "law of the minimum" remains one of the central concepts of nutrient limitation theory in biogeochemistry.

With the connection between plants and soils established, the concept of biogeochemistry gained more direction through the work of the founding fathers of soil science, including **Vasily Dokuchaev**, who linked the composition and creation of soil materials with the biota growing on them. Many of these scientists were interested in mineral composition and diagenesis, so their studies centered on the relationship between biota living in ecosystems with the mineral matter of the rocks and soils on which they were based. In 1875, **Eduard Suess** coined the term "biosphere" and gave us the idea that soils and their associated biota were really an intermediate in the flow of materials between the lithosphere, hydrosphere, and atmosphere. These global cycling concepts were reinforced by advancements in geology, including concepts of the rock cycle pioneered by **James Hutton** in the late 18th century, but only gaining wide acceptance almost 100 years later. It was also at the end of the 19th century that **Walther Nernst** established the thermodynamic principles of reduction and oxidation chemistry leading to the creation of the Nernst equation.

Almost at the same time in the later part of the 19th century, important discoveries were being made on a much, much smaller scale. These were the great advancements leading to the birth of modern microbial ecology. After discoveries that microorganisms were involved in the putrefaction of organic matter and fermentation, scientists like **Winogradsky** and **Beijerinck** properly tied microbial identity and function to global elemental cycles with the discovery of processes like nitrogen fixation, nitrification and denitrification, and sulfate reduction. These process discoveries were pivotal in completing our understanding of the cyclic nature of elements, and particularly that of nitrogen. Early work of **Waksman**, a biochemist and microbiologist, expanded on the application of microbial processes with studies on microorganisms and soil fertility, decomposition of plant and animal residues, formation of humus, and the occurrence and role of bacteria in marine processes.

The discoveries of microbial ecology were earth shattering, leading scientists to begin to see the cycles and spheres of the earth as having the ability to be sustainable or self-regulating. In the early 1900s, a mineralogist and geochemist, **Vladimir Vernadsky**, saw the importance of these types of processes as they related to the recycling of elements at the

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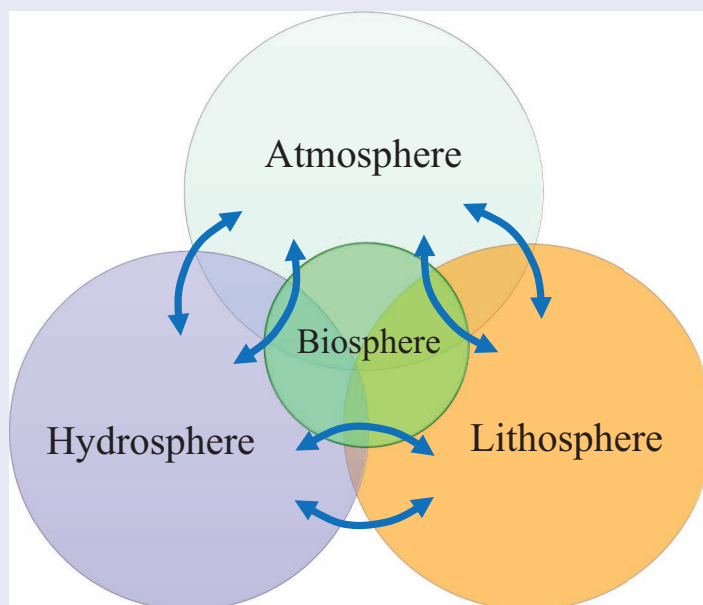


FIGURE 1.2 Conceptual diagram of the study of biogeochemistry as an interaction of the non-living earth spheres (lithosphere, hydrosphere, and atmosphere) with life in the biosphere.



FIGURE 1.3 Vladimir Vernadsky, the father of modern biogeochemistry, used the early disciplines of geochemistry and soil science to expand the concept of the biosphere and elemental cycling through biota.

ecosystem and global scales. His theories expanded on the biosphere concept of Suess, where he not only acknowledged that biota were “geologic force” but added that the biosphere was in unity with the cycling of elements through the other geologic spheres (the lithosphere, the atmosphere, and the hydrosphere). This idea is still at work today through concepts such as Gaia, where the earth is viewed as a living organism.

Because of his advancement of the biosphere concept and his original studies of element budgets in ecosystems, Vernadsky is often credited as the father of modern biogeochemistry. But it was not until a few more pieces of the puzzle were added that our modern view of biogeochemistry was truly realized. Following the acceptance of continental drift theory in the 1930s and 1940s, the final pieces of the modern biogeochemical puzzle were added with the emergence of ecology in the 1950s and 1960s with scientists like **Hutchinson** and **Redfield** and later **Odum**, who linked the populations of biota in ecosystems with their physical and chemical environments.

It was also in that time that the study of oxidation-reduction coupled to biogeochemical processes of paddy soils, lake and marine sediments, waterlogged, and poorly drained soils began to be appreciated by early scientists in the 1930s to 1950s, including **Sturgis**, **Pearsall**, **Mortimer**, **Patrick**, **Ponnamperuma**, and many others not included. The next generation of scientists in the early 1970s (including the authors of this book) and their subsequent discoveries have expanded the discipline with major wetland biogeochemistry research programs at several universities in the United States and in other countries. In the 1990s and through today, we have seen the emergence of wetland and aquatic systems biogeochemistry as a proper scientific discipline complete with dedicated journals, textbooks, and scientists willing to label themselves as biogeochemists.

Many of the problems being addressed by biogeochemistry are large in nature (e.g., ocean productivity, climate change), and some may hold an elitist view that biogeochemistry is only defined at the global scale. However, as demonstrated in this brief history of the concept, many scales are involved in the coupled cycling of elements, from microbes to continents. All of these related disciplines and their discoveries are to be credited with developing this notion to understand the interaction of life with the physical and chemical environments of the Earth.

Biogeochemical cycles are influenced by various processes that result in exchange of materials between two storage pools. The amount of a given constituent in these pools depends on its residence time, which is simply the amount of that material in the reservoir divided by the rate at which it is removed or added to the reservoir or the rate at which it is transformed. Living components in wetland soils can act as exchange or cycling pools, with rapid turnover and flow between the organisms in the pool and their immediate environment. The exchange or cycling pool can encompass up to 20% of the total amount of a given compound of a system and turn over rapidly, immobilizing and remobilizing compounds in a short time. Reservoir pools, which are larger with slower turnover, provide long-term storage. The reservoir pool typically contains the majority of a given compound in a system, is less reactive, and provides long-term storage. As an example, when wetlands are used for wastewater treatment, designs that increase the percentage of contaminants in the reservoir pool are more desirable because this provides long-term removal of the contaminant.

Wetland biogeochemistry involves processes by which an element or a compound is transformed within wetlands, including means by which various forms are interchanged between the solid, liquid, and gaseous phases. Thus, the broader ecosystem biogeochemistry definition is also applied to wetland biogeochemistry, which focuses on surface or near-surface processes in wetlands that govern biogeochemical cycles, plant production, microbial transformations, nutrient availability, pollutant removal, heavy metal chemistry, atmospheric exchange, and sediment transport. In this way, wetland biogeochemistry has its parentage in biology, soil science, chemistry, and geology and now takes its place along with the bio- and geosciences in providing the knowledge needed to develop solutions to the challenges faced by wetland and global ecosystems.

Biogeochemical processes in wetlands involve a host of complex microbial communities, periphyton, and vegetation, and interaction and mutual dependency among these communities. Biogeochemical processes also quantify the exchange and transport of elements or compounds within wetlands, including those to other systems (e.g., the atmosphere). Wetland biogeochemical processes provide overviews of the environmental factors and alterations in wetlands that control or affect functioning at the local, regional, and global level and can have both positive and negative feedbacks. Nitrogen and carbon cycles are good examples of biogeochemical processes that affect processes at the local level (e.g., plant growth and soil accretion, regional level (water quality), and the global level (greenhouse gas emissions and carbon storage). Wetland biogeochemical cycles can have global significance as follows (Figure 1.4):

- Eutrophication of oligotrophic wetlands resulting from increased nutrient loads can enhance primary productivity.
- High nutrient loads can affect species composition and food webs, leading to a dominance of fast-growing, high-reproduction organisms.
- High primary productivity can result in increased rates of organic matter accumulation, providing a sink for carbon (increased carbon sequestration).
- High rates of carbon accumulation can enhance microbial activities in soil and the water column, leading to increased nutrient regeneration and biotic growth.
- Increased rates of microbial activities can increase the production of greenhouse gases, and increased levels of greenhouse gases can have a negative impact on climate.

As we walk through a wetland, we all admire beautiful plants, flowers, birds and other wildlife, and flowing water, but we rarely think about the “living soil” under our feet. The biogeochemical

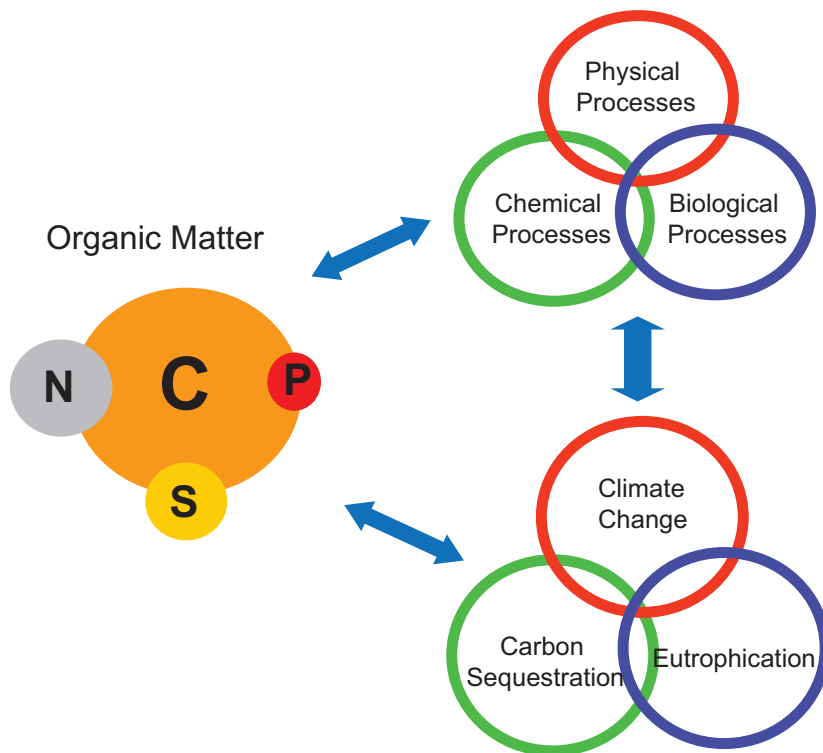


FIGURE 1.4 Linkages between physical, chemical, and biological processes and global-scale processes in the biosphere.

processes in the soil control many functions and ecosystem services provided by wetlands. This is similar to the “brain” orchestrating the many functions of the human body. The type of biogeochemical transformation occurring in wetlands, in contrast to upland systems, is strongly governed by hydrology. Transformations that occur in wetlands involve both anaerobic and aerobic processes. The chapters presented in this book focus on the role excess water and soil oxidation–reduction processes play in elemental cycling, heavy metal transformation, wetland plant response, and toxic organic transformation. The “**hub**” for biogeochemistry is organic matter, which contains macroelements including carbon, nitrogen, phosphorus, potassium, and sulfur (Figure 1.5). The cycling of these macroelements in the aerobic and anaerobic zones of the soil and water column is a primary driver supporting ecosystem processes and is far more interesting and important than any other constituent of wetlands.

Nutrients such as nitrogen, phosphorus, and sulfur are primary components of soil organic matter, and the cycling of these nutrients is always coupled with carbon cycling. The rate and extent of many of these reactions in the soil and the water column, involving carbon, nitrogen, phosphorus, and sulfur, are mediated by microbial communities and associated physicochemical reactions. These cycles are coupled, and there is considerable mutual dependency of one cycle on another (feedbacks and controls) or one organism on another (microbes, algae, and vegetation). There is a strong linkage between biogeochemical processes and biotic communities (vegetation, algae, and microbes). These coupled cycles operate at different spatial and temporal scales from the molecular to landscape level.

As you can see, wetlands and their biogeochemical processes are key drivers of several ecosystem functions associated with wetland values (e.g., water quality improvement through denitrification and long-term nutrient or carbon storage in the organic matter; Figure 1.5). Wetlands are

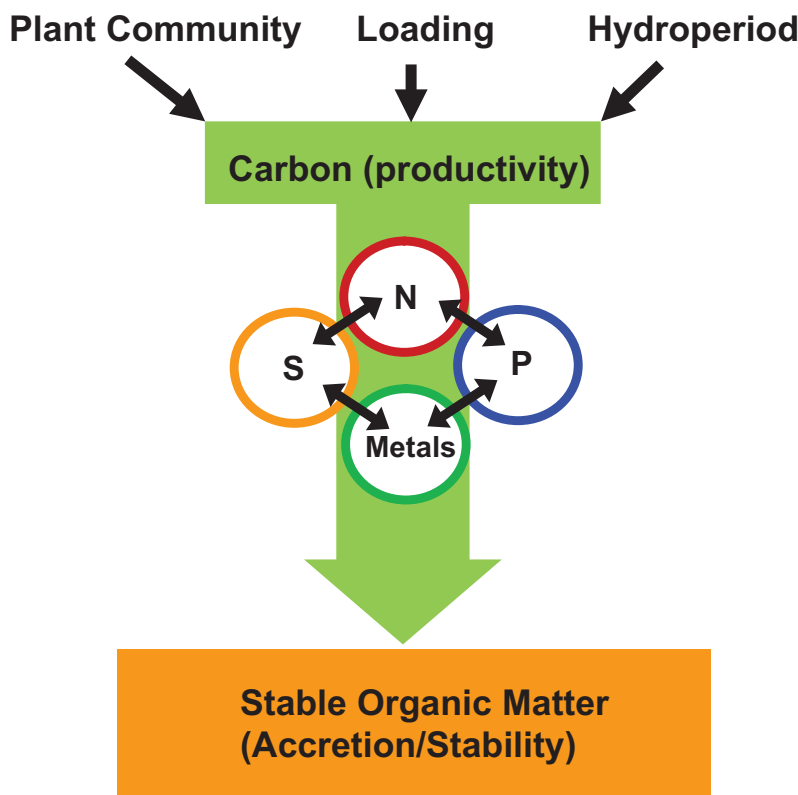


FIGURE 1.5 Relationship between coupled biogeochemical cycles and organic matter accretion in wetlands.

atmospheric sources of carbon dioxide, methane, and nitrous oxide, and due to flooded or reducing conditions, wetlands also limit organic matter turnover to serve as important global carbon sinks. Thus, biogeochemical processes in wetlands are also important in global issues, such as global warming, carbon sequestration, and eutrophication and water quality. For these reasons, knowledge of wetland biogeochemical processes is useful both for understanding the ecology of these systems leading to their preservation and to help us manage and predict the environmental fate and transport of elements and compounds from natural or anthropogenic sources. In both ways, the study of wetland biogeochemistry from the micro- to the macro-scale is of critical importance for the challenging issues we face today.

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